1 Synchronized divisor packets

2 Radical-excess obstructions to packet compression

Author: ChatGPT. We continue with the synchronized divisor packets of Section 1. The purpose of this section is twofold. First, we identify the exact order-statistic geometry hidden in the pair-max envelope. Second, we correct the proposed radical-scale packet gate by giving an infinite complete-premise counterfamily. The counterfamily disproves the packet gate, not the *abc* conjecture.

Let

$$R = \mathrm{rad}(abc), \qquad E = \frac{abc}{R}.$$

The integer E will be called the radical excess. Recall that every synchronized packet $Q = (x, y, z)$ satisfies $abc \le B(Q)$.

2.1 The shape identity

Put

$$m(Q) = \min(x, y, z), \qquad T(Q) = \max(x, y, z),$$

and

$$L(Q) = \max(y, z)\max(x, z)\max(x, y).$$

Thus $B(Q) = L(Q)^2$.

Theorem 2.1 (Symmetric packet-shape identity)**.** *For every synchronized packet,*

$$m(Q)L(Q) = xyzT(Q), \qquad m(Q)^2B(Q) = \big(xyzT(Q)\big)^2.$$

Proof. Sort the coordinates as $r \le s \le t$. Their pair maxima are s, t, t, so $m = r$, $T = t$, and $L = st^2$. Consequently $mL = rst^2 = (rst)t = xyzT$. Squaring proves the second identity. Since both sides are symmetric, the computation is independent of the sorting permutation. □

Proposition 2.2 (Support and full-envelope monotonicity)**.** *Every synchronized packet satisfies*

$$xyz \mid abc, \qquad \mathrm{rad}(xyz) \mid R, \qquad B(Q) \le B(a, b, c).$$

Proof. Multiply the three coordinate divisibilities. Radical divisibility is monotone under divisibility for nonzero natural numbers. Finally, $x \le a$, $y \le b$, and $z \le c$; hence every pair maximum for Q is no larger than its full-packet counterpart. Square and multiply. □

Equality of coordinate support with full radical support is false. The complete-premise packet $(5, 7, 2)$ over $(5, 7, 12)$ has coordinate radical 70, whereas $\mathrm{rad}(5 \cdot 7 \cdot 12) = 210$.

2.2 The radical-excess obstruction

Theorem 2.3 (Necessary condition for rational packet compression)**.** *Let $m, n \ge 0$. If*

$$B(Q)^m \le R^{m+n},$$

then

$$E^m \le R^n.$$

Equivalently, $R^n < E^m$ implies $R^{m+n} < B(Q)^m$ for every synchronized packet Q over the same primitive triple.

Proof. Since $R \mid abc$, one has $RE = abc$. Therefore

$$R^m E^m = (abc)^m \le B(Q)^m \le R^{m+n} = R^m R^n.$$

The radical is positive, so cancellation of R^m gives the assertion. $\square$

This condition depends only on the underlying triple. In particular, no choice inside the packet spectrum can repair its failure.

Theorem 2.4 (Infinite four-thirds obstruction)**.** *For $k \ge 0$, set*

$$A_k = 2^{k+4}, \qquad P_k = (A_k, 3, A_k + 3).$$

These are pairwise distinct primitive nonunit triples. If $R_k = \mathrm{rad}(A_k \cdot 3 \cdot (A_k + 3))$, then every synchronized packet over P_k satisfies

$$R_k^4 < B(Q)^3.$$

Consequently the locus on which every packet violates $B^3 \le R^4$ is infinite.

Proof. Put $h = 2^{k+3}$, so $A_k = 2h$ and $h \ge 8$. The triple is primitive because A_k is a power of 2 and is coprime to 3. Radical submultiplicativity gives

$$R_k \le \mathrm{rad}(A_k)\,\mathrm{rad}(3)\,\mathrm{rad}(A_k + 3) \le 2 \cdot 3 \cdot (A_k + 3) = 6(2h + 3).$$

Writing $E_k = A_k 3(A_k + 3)/R_k$, the equality $R_k E_k = 6h(A_k + 3)$ and the preceding upper bound imply $h \le E_k$. Moreover,

$$R_k \le 6(2h + 3) = 12h + 18 < 64h \le h^3 \le E_k^3.$$

Theorem 2.3, with $(m, n) = (3, 1)$, now gives $R_k^4 < B(Q)^3$ for every packet. The map $k \mapsto P_k$ is injective because its first coordinate is strictly increasing. $\square$

For positive real R and B, an estimate $B \le R^{4/3}$ implies $B^3 \le R^4$. Theorem 2.4 therefore refutes the earlier all-but-finitely-many packet gate $B(Q) \le R^{1+\varepsilon}$ at the exact value $\varepsilon = 1/3$. It does not produce triples violating $c \le R^{1+\varepsilon}$, and hence gives no counterexample to *abc*.

2.3 A correctly compensated forward gate

Theorem 2.5 (Compensated packet implication)**.** *Let $m, n \ge 0$. If*

$$B(Q)^m \le (ab)^m R^{m+n},$$

then

$$c^m \le R^{m+n}.$$

Proof. The synchronized product bound gives

$$(ab)^m c^m = (abc)^m \le B(Q)^m \le (ab)^m R^{m+n}.$$

Cancel the positive factor $(ab)^m$. $\square$

Thus the factor ab is the exact elementary compensator that changes a packet bound for the full product into the desired bound for the sum arm. No eventual compensated estimate is proved here. Its pointwise four-thirds form already fails at $(2, 3, 5)$, whose packet spectrum consists only of the full packet. The all-but-finitely-many compensated direction remains open because this section supplies no infinite counterfamily to it.

2.4 Formal and computational audit

The Lean module `SynchronizedPacketRadicalExcessObstruction20260903.lean` formalizes the shape identity, support containment, full-envelope monotonicity, the radical-excess implication, the dyadic infinite obstruction locus, the compensated implication, and the complete-premise finite counterexamples. Its axiom audit contains only `propext`, `Classical.choice`, and `Quot.sound`.

An exact integer search exhausts every normalized primitive nonunit triple with $c \leq 3000$ and every divisor triple satisfying all packet premises. It checks 1,365,095 primitive triples, 1,366,531 packets, and 1,436 proper packets. The same program separately factors and exhausts the first twenty-one dyadic rows. Null finite searches are recorded only as evidence and are not promoted to universal statements.